

# Face Recognition Using PCA and Artificial Neural Network (ANN)

## Abstract

Face recognition is one of the most widely used applications of machine learning and computer vision. This project presents a face recognition system based on Principal Component Analysis (PCA) and Artificial Neural Networks (ANN). PCA is used to extract important facial features and reduce dimensionality, while ANN performs classification and recognition of faces. The proposed system was evaluated using a face image dataset and achieved a recognition accuracy of 61%.

## Introduction

Face recognition is a biometric technique used to identify or verify individuals based on facial characteristics. Traditional image data contains a large number of features, which increases computational complexity. PCA helps reduce dimensionality by selecting the most informative features. ANN then learns patterns from these extracted features and performs classification.

## Objectives

- Develop a face recognition system using PCA and ANN.
- Reduce dimensionality of face images using Eigenfaces.
- Train an ANN classifier for face identification.
- Evaluate system performance using classification accuracy.
- Analyze the effect of varying PCA components on recognition performance.

## Methodology

### Data Preprocessing

- Face images were loaded from the dataset.
- Images were converted to grayscale.
- Images were resized to a fixed size.
- Pixel values were normalized between 0 and 1.

### Train-Test Split

The dataset was divided into:

- Training Data: 60%
- Testing Data: 40%

## Principal Component Analysis (PCA)

PCA was applied to:

- Reduce dimensionality.
- Extract significant facial features.
- Generate Eigenfaces for feature representation.

## Artificial Neural Network

The ANN architecture consisted of:

- Input Layer
- Hidden Dense Layers
- Dropout Layers
- Output Layer with Softmax Activation

The model was trained using the Adam optimizer and categorical cross-entropy loss function.

## Results

The trained model was evaluated on the testing dataset.

**Final Test Accuracy: 61%**

An Accuracy vs K analysis was also performed to study the impact of the number of PCA components on classification performance.

## Advantages

- Reduced computational complexity.
- Efficient feature extraction using PCA.
- Improved learning through ANN.
- Easy implementation using Python libraries.

## Limitations

- Accuracy depends on image quality.
- Performance may decrease with large variations in pose and illumination.
- Recognition accuracy can be improved using larger datasets and deeper neural networks.

## Conclusion

A face recognition system based on PCA and ANN was successfully implemented. PCA effectively reduced dimensionality and extracted relevant facial features, while ANN performed classification using the extracted features. The model achieved a maximum testing

accuracy of 61%, demonstrating the effectiveness of combining PCA and ANN for face recognition tasks.

## **Future Scope**

- Use Convolutional Neural Networks (CNNs) for higher accuracy.
- Train on larger and more diverse face datasets.
- Implement real-time face recognition using webcam input.
- Explore advanced feature extraction techniques.